

## Homework 2: Lambda Calculus — Solutions

### Problem 1

Let

$$I = \lambda x. x, \quad R = I I, \quad B = \lambda f. (\lambda u. f) (f I).$$

Normal-order evaluation takes three steps; eager evaluation takes four:

$$\begin{aligned} B R &\rightarrow_{\text{NO}} (\lambda u. R) (R I) \rightarrow_{\text{NO}} R \rightarrow_{\text{NO}} I, \\ B R &\rightarrow_{\text{E}} B I \rightarrow_{\text{E}} (\lambda u. I) (I I) \rightarrow_{\text{E}} (\lambda u. I) I \rightarrow_{\text{E}} I. \end{aligned}$$

Both return  $I$ , which is both a canonical form and a normal form.

Normal-order big-step derivation:

$$\frac{\overline{B \Rightarrow_{\text{NO}} B} \quad \frac{\overline{\lambda u. R \Rightarrow_{\text{NO}} \lambda u. R} \quad \frac{\overline{I \Rightarrow_{\text{NO}} I} \quad \overline{I \Rightarrow_{\text{NO}} I}}{\overline{R \Rightarrow_{\text{NO}} I}}}{\overline{(\lambda u. R) (R I) \Rightarrow_{\text{NO}} I}}}{\overline{B R \Rightarrow_{\text{NO}} I}}$$

Eager big-step derivation:

$$\frac{\overline{B \Rightarrow_{\text{E}} B} \quad \frac{\overline{I \Rightarrow_{\text{E}} I} \quad \overline{I \Rightarrow_{\text{E}} I} \quad \overline{I \Rightarrow_{\text{E}} I}}{\overline{R \Rightarrow_{\text{E}} I}} \quad \frac{\overline{\lambda u. I \Rightarrow_{\text{E}} \lambda u. I} \quad \frac{\overline{I \Rightarrow_{\text{E}} I} \quad \overline{I \Rightarrow_{\text{E}} I} \quad \overline{I \Rightarrow_{\text{E}} I}}{\overline{I I \Rightarrow_{\text{E}} I}} \quad \overline{I \Rightarrow_{\text{E}} I}}{\overline{(\lambda u. I) (I I) \Rightarrow_{\text{E}} I}}}{\overline{B R \Rightarrow_{\text{E}} I}}$$

**Problem 1: alternative big-step derivations.** The following uses the indented style from the lecture notes. For each application, its premises are indented below it; the result is then shown at the application's indentation level. All terms are written out in full.

Normal-order evaluation:

$$\begin{aligned}
& (\lambda f. (\lambda u. f) (f (\lambda x. x))) ((\lambda x. x)(\lambda x. x)) \\
& \quad \lambda f. (\lambda u. f) (f (\lambda x. x)) \Rightarrow_{\text{NO}} \lambda f. (\lambda u. f) (f (\lambda x. x)) \\
& \quad (\lambda u. (\lambda x. x)(\lambda x. x)) (((\lambda x. x)(\lambda x. x))(\lambda x. x)) \\
& \quad \quad \lambda u. (\lambda x. x)(\lambda x. x) \Rightarrow_{\text{NO}} \lambda u. (\lambda x. x)(\lambda x. x) \\
& \quad \quad (\lambda x. x)(\lambda x. x) \\
& \quad \quad \quad \lambda x. x \Rightarrow_{\text{NO}} \lambda x. x \\
& \quad \quad \quad \lambda x. x \Rightarrow_{\text{NO}} \lambda x. x \\
& \quad \quad \Rightarrow_{\text{NO}} \lambda x. x \\
& \quad \Rightarrow_{\text{NO}} \lambda x. x \\
& \Rightarrow_{\text{NO}} \lambda x. x
\end{aligned}$$

Eager evaluation:

$$\begin{aligned}
& (\lambda f. (\lambda u. f) (f (\lambda x. x))) ((\lambda x. x)(\lambda x. x)) \\
& \quad \lambda f. (\lambda u. f) (f (\lambda x. x)) \Rightarrow_{\text{E}} \lambda f. (\lambda u. f) (f (\lambda x. x)) \\
& \quad (\lambda x. x)(\lambda x. x) \\
& \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \Rightarrow_{\text{E}} \lambda x. x \\
& \quad (\lambda u. \lambda x. x) ((\lambda x. x)(\lambda x. x)) \\
& \quad \quad \lambda u. \lambda x. x \Rightarrow_{\text{E}} \lambda u. \lambda x. x \\
& \quad \quad (\lambda x. x)(\lambda x. x) \\
& \quad \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \quad \lambda x. x \Rightarrow_{\text{E}} \lambda x. x \\
& \quad \Rightarrow_{\text{E}} \lambda x. x \\
& \Rightarrow_{\text{E}} \lambda x. x
\end{aligned}$$

## Problem 2

Let

$$W = \lambda v. (\lambda w. w) v, \quad F = \lambda u. W, \quad \Delta = \lambda x. x x, \quad \Omega = \Delta \Delta.$$

Normal-order evaluation terminates:

$$F \Omega \rightarrow_{\text{NO}} W, \quad \frac{\overline{F \Rightarrow_{\text{NO}} F} \quad \overline{W \Rightarrow_{\text{NO}} W}}{F \Omega \Rightarrow_{\text{NO}} W}$$

The result  $W = \lambda v. (\lambda w. w) v$  is a canonical form, but not a normal form, since  $(\lambda w. w) v$  is a redex.

Eager evaluation diverges:

$$F \Omega \rightarrow_{\text{E}} F \Omega \rightarrow_{\text{E}} \dots$$

There is no finite big-step derivation for this divergent evaluation.

### Alternative: indented big-step evaluation.

Normal-order evaluation:

$$\begin{aligned} & (\lambda u. \lambda v. (\lambda w. w) v) ((\lambda x. x x)(\lambda x. x x)) \\ & \quad \lambda u. \lambda v. (\lambda w. w) v \Rightarrow_{\text{NO}} \lambda u. \lambda v. (\lambda w. w) v \\ & \quad \lambda v. (\lambda w. w) v \Rightarrow_{\text{NO}} \lambda v. (\lambda w. w) v \\ & \Rightarrow_{\text{NO}} \lambda v. (\lambda w. w) v \end{aligned}$$

The argument is not evaluated. Substituting it for  $u$  leaves  $\lambda v. (\lambda w. w) v$  unchanged, and evaluation stops at this abstraction.

### Problem 3

(a) For the first term, translate each abstraction and application:

$$\begin{aligned}
& \mathcal{D}_{()}(\lambda x. \lambda y. x (\lambda x. y x)) \\
&= \lambda. \mathcal{D}_{(x)}(\lambda y. x (\lambda x. y x)) \\
&= \lambda. \lambda. \mathcal{D}_{(x,y)}(x (\lambda x. y x)) \\
&= \lambda. \lambda. (\mathcal{D}_{(x,y)}(x) \mathcal{D}_{(x,y)}(\lambda x. y x)) \\
&= \lambda. \lambda. (\mathcal{D}_{(x,y)}(x) (\lambda. \mathcal{D}_{(x,y,x)}(y x))) \\
&= \lambda. \lambda. (\mathcal{D}_{(x,y)}(x) (\lambda. (\mathcal{D}_{(x,y,x)}(y) \mathcal{D}_{(x,y,x)}(x)))) \\
&= \lambda. \lambda. (1 (\lambda. (1 0))).
\end{aligned}$$

Here the variable translations use the rightmost occurrence of each name:

$$\begin{aligned}
\mathcal{D}_{(x,y)}(x) &= (x, y)(x) = 1 + (x)(x) = 1 + 0 = 1, \\
\mathcal{D}_{(x,y,x)}(y) &= (x, y, x)(y) = 1 + (x, y)(y) = 1 + 0 = 1, \\
\mathcal{D}_{(x,y,x)}(x) &= (x, y, x)(x) = 0.
\end{aligned}$$

For the second term:

$$\begin{aligned}
& \mathcal{D}_{()}((\lambda x. \lambda x. x)(\lambda y. y)) \\
&= \mathcal{D}_{()}(\lambda x. \lambda x. x) \mathcal{D}_{()}(\lambda y. y) \\
&= (\lambda. \mathcal{D}_{(x)}(\lambda x. x)) \mathcal{D}_{()}(\lambda y. y) \\
&= (\lambda. \lambda. \mathcal{D}_{(x,x)}(x)) \mathcal{D}_{()}(\lambda y. y) \\
&= (\lambda. \lambda. \mathcal{D}_{(x,x)}(x)) (\lambda. \mathcal{D}_{(y)}(y)) \\
&= (\lambda. \lambda. 0)(\lambda. 0).
\end{aligned}$$

Here the variable translations are:

$$\mathcal{D}_{(x,x)}(x) = (x, x)(x) = 0, \quad \mathcal{D}_{(y)}(y) = (y)(y) = 0.$$

(b) The cutoff increases under each abstraction; indices below the cutoff stay unchanged.

(i)

$$\begin{aligned}
\uparrow_0^2 (\lambda. (1 (\lambda. (2 \ 0)))) &= \lambda. (\uparrow_1^2 (1 (\lambda. (2 \ 0)))) \\
&= \lambda. ((\uparrow_1^2 1) (\uparrow_1^2 (\lambda. (2 \ 0)))) \\
&= \lambda. (3 (\uparrow_1^2 (\lambda. (2 \ 0)))) \\
&= \lambda. (3 (\lambda. \uparrow_2^2 (2 \ 0))) \\
&= \lambda. (3 (\lambda. ((\uparrow_2^2 2) (\uparrow_2^2 0)))) \\
&= \lambda. (3 (\lambda. (4 (\uparrow_2^2 0)))) \\
&= \lambda. (3 (\lambda. (4 \ 0))).
\end{aligned}$$

(ii)

$$\begin{aligned}
\uparrow_0^{-1} (\lambda. (2 (\lambda. (3 \ 0)))) &= \lambda. (\uparrow_1^{-1} (2 (\lambda. (3 \ 0)))) \\
&= \lambda. ((\uparrow_1^{-1} 2) (\uparrow_1^{-1} (\lambda. (3 \ 0)))) \\
&= \lambda. (1 (\uparrow_1^{-1} (\lambda. (3 \ 0)))) \\
&= \lambda. (1 (\lambda. \uparrow_2^{-1} (3 \ 0))) \\
&= \lambda. (1 (\lambda. ((\uparrow_2^{-1} 3) (\uparrow_2^{-1} 0)))) \\
&= \lambda. (1 (\lambda. (2 (\uparrow_2^{-1} 0)))) \\
&= \lambda. (1 (\lambda. (2 \ 0))).
\end{aligned}$$

(iii) Let  $s = \lambda. (1 \ 0)$ . The two shifts of the replacement term are:

$$\begin{array}{ll}
s_1 = \uparrow_0^1 s & s_2 = \uparrow_0^1 s_1 \\
= \uparrow_0^1 (\lambda. (1 \ 0)) & = \uparrow_0^1 (\lambda. (2 \ 0)) \\
= \lambda. (\uparrow_1^1 (1 \ 0)) & = \lambda. (\uparrow_1^1 (2 \ 0)) \\
= \lambda. ((\uparrow_1^1 1) (\uparrow_1^1 0)) & = \lambda. ((\uparrow_1^1 2) (\uparrow_1^1 0)) \\
= \lambda. (2 (\uparrow_1^1 0)) & = \lambda. (3 (\uparrow_1^1 0)) \\
= \lambda. (2 \ 0), & = \lambda. (3 \ 0).
\end{array}$$

Applying the substitution rules gives

$$\begin{aligned}
&(\lambda. (1 (\lambda. (2 \ 0))))[0 := s] \\
&= \lambda. ((1 (\lambda. (2 \ 0)))[1 := \uparrow_0^1 s]) \\
&= \lambda. ((1 (\lambda. (2 \ 0)))[1 := s_1]) \\
&= \lambda. ((1[1 := s_1]) ((\lambda. (2 \ 0))[1 := s_1])) \\
&= \lambda. (s_1 ((\lambda. (2 \ 0))[1 := s_1])) \\
&= \lambda. (s_1 (\lambda. ((2 \ 0)[2 := \uparrow_0^1 s_1]))) \\
&= \lambda. (s_1 (\lambda. ((2 \ 0)[2 := s_2]))) \\
&= \lambda. (s_1 (\lambda. ((2[2 := s_2]) (0[2 := s_2])))) \\
&= \lambda. (s_1 (\lambda. (s_2 (0[2 := s_2])))) \\
&= \lambda. (s_1 (\lambda. (s_2 \ 0))) \\
&= \lambda. ((\lambda. (2 \ 0)) (\lambda. ((\lambda. (3 \ 0)) 0))).
\end{aligned}$$

(c) Each  $\rightarrow$  applies the reduction rules; each  $=$  unfolds the shifting or substitution definitions, or an abbreviation. Shifting and substitution are operations on terms, so computing them adds no reduction steps.

(i) **Nameless reduction.**

$$\begin{aligned}
& \lambda. ((\lambda. \lambda. (1 \ 0)) \ 0) \\
\rightarrow & \lambda. \uparrow_0^{-1} ((\lambda. (1 \ 0)) [0 := \uparrow_0^1 0]) \\
= & \lambda. \uparrow_0^{-1} ((\lambda. (1 \ 0)) [0 := 1]) \\
= & \lambda. \uparrow_0^{-1} (\lambda. ((1 \ 0) [1 := \uparrow_0^1 1])) \\
= & \lambda. \uparrow_0^{-1} (\lambda. ((1 \ 0) [1 := 2])) \\
= & \lambda. \uparrow_0^{-1} (\lambda. ((1 [1 := 2]) (0 [1 := 2]))) \\
= & \lambda. \uparrow_0^{-1} (\lambda. (2 (0 [1 := 2]))) \\
= & \lambda. \uparrow_0^{-1} (\lambda. (2 \ 0)) \\
= & \lambda. \lambda. \uparrow_1^{-1} (2 \ 0) \\
= & \lambda. \lambda. ((\uparrow_1^{-1} 2) (\uparrow_1^{-1} 0)) \\
= & \lambda. \lambda. (1 (\uparrow_1^{-1} 0)) \\
= & \lambda. \lambda. (1 \ 0).
\end{aligned}$$

**Named reduction.** A corresponding named term reduces in the same order:

$$\lambda a. ((\lambda u. \lambda b. u \ b) \ a) \rightarrow \lambda a. \lambda b. a \ b.$$

The final translation agrees with the nameless normal form:

$$\mathcal{D}_\emptyset(\lambda a. \lambda b. a \ b) = \lambda. \lambda. (1 \ 0).$$

(ii) **Nameless reduction.** Let  $I = \lambda. 0$  and  $T = \lambda. \lambda. (1 (1 0))$ . The shifts of  $I$  used below are

$$\begin{aligned}\uparrow_0^1 I &= \uparrow_0^1 (\lambda. 0) = \lambda. (\uparrow_1^1 0) = \lambda. 0 = I, \\ \uparrow_1^{-1} I &= \uparrow_1^{-1} (\lambda. 0) = \lambda. (\uparrow_2^{-1} 0) = \lambda. 0 = I.\end{aligned}$$

The reduction takes three steps:

$$\begin{aligned}T I &\rightarrow \uparrow_0^{-1} ((\lambda. (1 (1 0))) [0 := \uparrow_0^1 I]) \\ &= \uparrow_0^{-1} ((\lambda. (1 (1 0))) [0 := I]) \\ &= \uparrow_0^{-1} (\lambda. ((1 (1 0)) [1 := \uparrow_0^1 I])) \\ &= \uparrow_0^{-1} (\lambda. ((1 (1 0)) [1 := I])) \\ &= \uparrow_0^{-1} (\lambda. ((1 [1 := I]) ((1 0) [1 := I]))) \\ &= \uparrow_0^{-1} (\lambda. (I ((1 0) [1 := I]))) \\ &= \uparrow_0^{-1} (\lambda. (I ((1 [1 := I]) (0 [1 := I]))) \\ &= \uparrow_0^{-1} (\lambda. (I (I (0 [1 := I]))) \\ &= \uparrow_0^{-1} (\lambda. (I (I 0))) \\ &= \lambda. \uparrow_1^{-1} (I (I 0)) \\ &= \lambda. ((\uparrow_1^{-1} I) (\uparrow_1^{-1} (I 0))) \\ &= \lambda. (I (\uparrow_1^{-1} (I 0))) \\ &= \lambda. (I ((\uparrow_1^{-1} I) (\uparrow_1^{-1} 0))) \\ &= \lambda. (I (I (\uparrow_1^{-1} 0))) \\ &= \lambda. (I (I 0)) \\ &\rightarrow \lambda. (I (\uparrow_0^{-1} (0 [0 := \uparrow_0^1 0]))) \\ &= \lambda. (I (\uparrow_0^{-1} (0 [0 := 1]))) \\ &= \lambda. (I (\uparrow_0^{-1} 1)) \\ &= \lambda. (I 0) \\ &\rightarrow \lambda. \uparrow_0^{-1} (0 [0 := \uparrow_0^1 0]) \\ &= \lambda. \uparrow_0^{-1} (0 [0 := 1]) \\ &= \lambda. \uparrow_0^{-1} 1 \\ &= \lambda. 0.\end{aligned}$$

**Named reduction.** A corresponding named term reduces in the same order:

$$\begin{aligned}&(\lambda f. \lambda x. f (f x)) (\lambda y. y) \\ &\rightarrow \lambda x. (\lambda y. y) ((\lambda y. y) x) \\ &\rightarrow \lambda x. (\lambda y. y) x \\ &\rightarrow \lambda x. x.\end{aligned}$$

The final translation agrees with the nameless normal form:

$$\mathcal{D}_0(\lambda x. x) = \lambda. 0.$$

(iii) **Nameless reduction.** Let

$$T = \lambda. \lambda. (1 (1 0)), \quad s = \lambda. (1 0).$$

The given term is  $\lambda. (T s)$ . The shifts used below are

$$\begin{array}{lll} s_1 = \uparrow_0^1 s & s_2 = \uparrow_0^1 s_1 & \uparrow_1^{-1} s_2 = \uparrow_1^{-1} (\lambda. (3 0)) \\ = \uparrow_0^1 (\lambda. (1 0)) & = \uparrow_0^1 (\lambda. (2 0)) & = \lambda. \uparrow_2^{-1} (3 0) \\ = \lambda. (\uparrow_1^1 (1 0)) & = \lambda. (\uparrow_1^1 (2 0)) & = \lambda. ((\uparrow_2^{-1} 3) (\uparrow_2^{-1} 0)) \\ = \lambda. ((\uparrow_1^1 1) (\uparrow_1^1 0)) & = \lambda. ((\uparrow_1^1 2) (\uparrow_1^1 0)) & = \lambda. (2 (\uparrow_2^{-1} 0)) \\ = \lambda. (2 (\uparrow_1^1 0)) & = \lambda. (3 (\uparrow_1^1 0)) & = \lambda. (2 0) \\ = \lambda. (2 0), & = \lambda. (3 0), & = s_1. \end{array}$$

The reduction takes three steps:

$$\begin{aligned} \lambda. (T s) &\rightarrow \lambda. \uparrow_0^{-1} ((\lambda. (1 (1 0))) [0 := \uparrow_0^1 s]) \\ &= \lambda. \uparrow_0^{-1} ((\lambda. (1 (1 0))) [0 := s_1]) \\ &= \lambda. \uparrow_0^{-1} (\lambda. ((1 (1 0)) [1 := \uparrow_0^1 s_1])) \\ &= \lambda. \uparrow_0^{-1} (\lambda. ((1 (1 0)) [1 := s_2])) \\ &= \lambda. \uparrow_0^{-1} (\lambda. ((1 [1 := s_2]) ((1 0) [1 := s_2]))) \\ &= \lambda. \uparrow_0^{-1} (\lambda. (s_2 ((1 0) [1 := s_2]))) \\ &= \lambda. \uparrow_0^{-1} (\lambda. (s_2 ((1 [1 := s_2]) (0 [1 := s_2]))) \\ &= \lambda. \uparrow_0^{-1} (\lambda. (s_2 (s_2 (0 [1 := s_2]))) \\ &= \lambda. \uparrow_0^{-1} (\lambda. (s_2 (s_2 0))) \\ &= \lambda. \lambda. \uparrow_1^{-1} (s_2 (s_2 0)) \\ &= \lambda. \lambda. ((\uparrow_1^{-1} s_2) (\uparrow_1^{-1} (s_2 0))) \\ &= \lambda. \lambda. (s_1 (\uparrow_1^{-1} (s_2 0))) \\ &= \lambda. \lambda. (s_1 ((\uparrow_1^{-1} s_2) (\uparrow_1^{-1} 0))) \\ &= \lambda. \lambda. (s_1 (s_1 (\uparrow_1^{-1} 0))) \\ &= \lambda. \lambda. (s_1 (s_1 0)) \\ &\rightarrow \lambda. \lambda. (s_1 (\uparrow_0^{-1} ((2 0) [0 := \uparrow_0^1 0]))) \\ &= \lambda. \lambda. (s_1 (\uparrow_0^{-1} ((2 0) [0 := 1]))) \\ &= \lambda. \lambda. (s_1 (\uparrow_0^{-1} ((2 [0 := 1]) (0 [0 := 1])))) \\ &= \lambda. \lambda. (s_1 (\uparrow_0^{-1} (2 (0 [0 := 1])))) \\ &= \lambda. \lambda. (s_1 (\uparrow_0^{-1} (2 1))) \\ &= \lambda. \lambda. (s_1 ((\uparrow_0^{-1} 2) (\uparrow_0^{-1} 1))) \\ &= \lambda. \lambda. (s_1 (1 (\uparrow_0^{-1} 1))) \\ &= \lambda. \lambda. (s_1 (1 0)). \end{aligned}$$

(To be continued on the next page.)

(iii), continued:

$$\begin{aligned}
& \lambda. \lambda. (s_1 (1 \ 0)) \\
\rightarrow & \lambda. \lambda. \uparrow_0^{-1} ((2 \ 0)[0 := \uparrow_0^1 (1 \ 0)]) \\
= & \lambda. \lambda. \uparrow_0^{-1} ((2 \ 0)[0 := ((\uparrow_0^1 1) (\uparrow_0^1 0))]) \\
= & \lambda. \lambda. \uparrow_0^{-1} ((2 \ 0)[0 := (2 (\uparrow_0^1 0))]) \\
= & \lambda. \lambda. \uparrow_0^{-1} ((2 \ 0)[0 := (2 \ 1)]) \\
= & \lambda. \lambda. \uparrow_0^{-1} ((2[0 := (2 \ 1)]) (0[0 := (2 \ 1)])) \\
= & \lambda. \lambda. \uparrow_0^{-1} (2 (0[0 := (2 \ 1)])) \\
= & \lambda. \lambda. \uparrow_0^{-1} (2 (2 \ 1)) \\
= & \lambda. \lambda. ((\uparrow_0^{-1} 2) (\uparrow_0^{-1} (2 \ 1))) \\
= & \lambda. \lambda. (1 (\uparrow_0^{-1} (2 \ 1))) \\
= & \lambda. \lambda. (1 ((\uparrow_0^{-1} 2) (\uparrow_0^{-1} 1))) \\
= & \lambda. \lambda. (1 (1 (\uparrow_0^{-1} 1))) \\
= & \lambda. \lambda. (1 (1 \ 0)).
\end{aligned}$$

**Named reduction.** A corresponding named term reduces in the same order:

$$\begin{aligned}
& \lambda a. ((\lambda f. \lambda b. f (f \ b)) (\lambda z. a \ z)) \\
\rightarrow & \lambda a. \lambda b. (\lambda z. a \ z) ((\lambda z. a \ z) \ b) \\
\rightarrow & \lambda a. \lambda b. (\lambda z. a \ z) (a \ b) \\
\rightarrow & \lambda a. \lambda b. a (a \ b).
\end{aligned}$$

The final translation agrees with the nameless normal form:

$$\mathcal{D}_0(\lambda a. \lambda b. a (a \ b)) = \lambda. \lambda. (1 (1 \ 0)).$$

(d)

$$\begin{aligned}
& \forall \Gamma, M, N, x. fv(M) \cup fv(N) \cup \{x\} \subseteq \text{dom}(\Gamma) \\
\implies & \mathcal{D}_\Gamma(M[N/x]) = \mathcal{D}_\Gamma(M)[\Gamma(x) := \mathcal{D}_\Gamma(N)].
\end{aligned}$$

The premise ensures that  $\mathcal{D}_\Gamma(M)$ ,  $\mathcal{D}_\Gamma(N)$ , and  $\Gamma(x)$  are defined; without it, the equality may contain undefined expressions.

**Alternative using shifts.**

$$\begin{aligned}
& \forall \Gamma, M, N, x. (fv(M) \setminus \{x\}) \cup fv(N) \subseteq \text{dom}(\Gamma) \\
\implies & \mathcal{D}_\Gamma(M[N/x]) = \uparrow_0^{-1} (\mathcal{D}_{\Gamma, x}(M)[0 := \uparrow_0^1 \mathcal{D}_\Gamma(N)]).
\end{aligned}$$

The shifts account for adding and removing the context entry for  $x$ . Both versions agree under the original premise, but this version also permits  $x \notin \text{dom}(\Gamma)$ .